



Scaling Laws Revisited

Modeling the Role of Data Quality in Language Model Pretraining

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[arXiv:2510.03313](https://arxiv.org/abs/2510.03313)

Scaling Law

Chinchilla scaling Law

N

D

Token

Chinchilla Scaling Law

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E$$

= + +

- L =
- N =
- D = Token
- A, B, E, α , β =

Token

?

-
- ---
- FineWeb Dolma
- Scaling Law " "

Scaling Law

Scaling Law

OCR

VS

- Bansal et al. (2022): NMT
- Goyal et al. (2024): $\neq$
- Chang et al. (2024):
- Chen et al. (2025):

Scaling law

Gap

"

"

Q

Chinchilla

□□□ $L(N, D, Q) = ?$ — □□□□□□□□□□□□

11

Q



Scaling Law

仅需在 Chinchilla 分母中新增一项 Q^ν ，就可以在 Scaling Law 中引入数据质量

Scaling Law

$$L(N, D, Q) = \frac{A}{N^\alpha} + \frac{B}{D^\beta Q^\nu} + E$$

= + +

Chinchilla

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E$$

N D

$$L(N, D, Q) = \frac{A}{N^\alpha} + \frac{B}{D^\beta Q^\nu} + E$$

$N + D + Q \rightarrow$

有效样本量 $D_{eff} = D \cdot Q^\nu$: 质量降级等价于有效样本量按 Q^ν 倍缩减

$\gamma \qquad \qquad \gamma \qquad \rightarrow \qquad \qquad \gamma \qquad \rightarrow$

Fisher

两种理论路径，共同推导出 $D_{eff} = D \cdot Q^\gamma \rightarrow$ 质量感知 Scaling Law

1

观测模型: $y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$

每个观测值的对数似然为: $l(\theta) = -\frac{1}{2\sigma^2} (y - \theta)^2 + c$

每个样本 Fisher 信息: $I_1(\theta) = \mathbb{E}[-\frac{\partial^2}{\partial \theta^2} l(\theta)] = 1/\sigma^2$

D 个样本总信息: $I_D = D/\sigma^2$

定义缺陷 $\Delta = \ln(\sigma^2/\sigma_0^2)$, $Q = e^{-\Delta} = \sigma_0^2/\sigma^2$

$$D_{eff} = D(\sigma_0^2/\sigma^2) = D \cdot Q^\gamma \quad (\gamma = 1)$$

2

二分类任务, 标签翻转率: $\eta \in [0, 1/2)$

如果用带噪声校正的损失函数来训练, 虽然可以得到无偏估计

但是代价是每个样本的方差会膨胀, 膨胀因子: $(1 - 2\eta)^{-2}$

$Q = 1 - \eta$, 有效样本量:

$$D_{eff} = D \cdot (2Q - 1)^2 \approx D \cdot Q^\gamma \quad (\gamma = 2)$$

1

—

+

- 设 $I(\tilde{X}; Z) = \rho(Q) \cdot I(X; Z)$ (噪声信道 C_Q 将互信息按 $\rho(Q)$ 比例缩减)
- 损失遵循: $L_D \propto (D \cdot I(\tilde{X}; Z))^{-\beta}$
- Scaling law

× × × = 126

NMT

- → ParaCrawl v8 (101M)
- 8 GPT Neo 133M
- BPE Tokenizer vocab size = 32000
- 0.5M / 1M / 2M

CLM

- C4 (en)
- 8 Llama 3 hidden=512
- Llama 3.2-1B Tokenizer
- 100M / 1B / 10B Tokens

2 × 3 × 7 × 3 = 126

× × ×

7

Q=1.0	Q=0.9	Q=0.8	Q=0.75	Q=0.7	Q=0.6	Q=0.5	$\eta=0\% \rightarrow 50\%$
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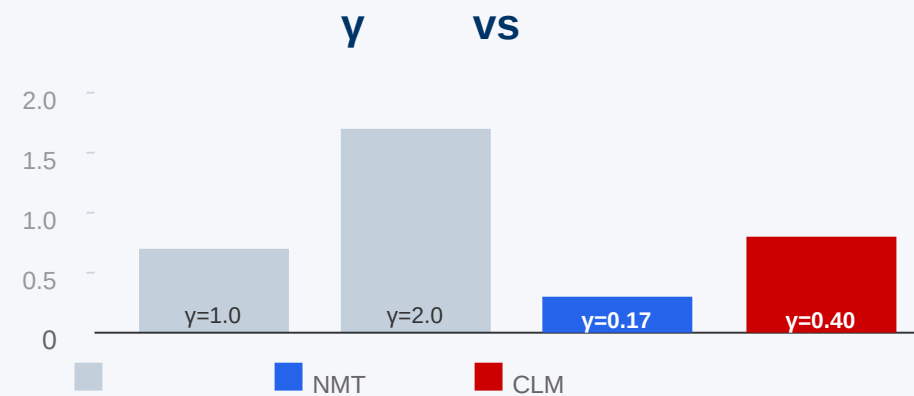
NMT 50% token padding CLM 15% token
嵌套子集策略: $S_{0.1B} \subseteq S_{1B} \subseteq S_{10B}$ (保证数据分布一致性)

γ

$\gamma \geq 1$ vs $\gamma \approx 0.17 \sim 0.40$ —

Scaling Law

$\square\square$	$\square\square$	β	γ
NMT	Hube	0.25	0.17
CLM	Hube	0.39	0.40
	1		1.0
	2		2.0



$\gamma \ll 1$

$y \ll 1$

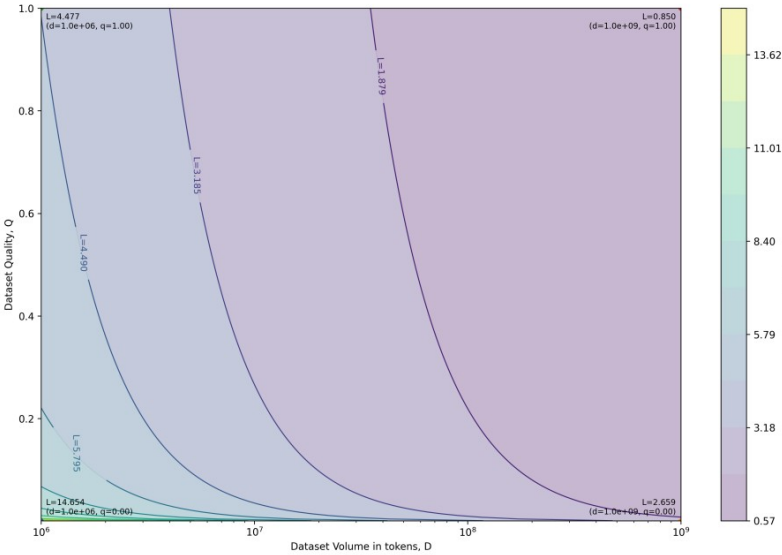
— Token

$y \ll 1$

- —
- 30% token
" "
- PAC /

⚡ NMT vs CLM

- NMT ($y=0.17,$)
- - " "
- CLM ($y=0.40,$)
- Token
 -



iso-loss vs
Q D → " Q "

Q

γ

$\gamma \approx 0.39$
($lr=5e-3$)

CLM

($lr=1e-5$)

$\Delta L(Q)$

$L(Q)-L(1)$ 对 $Q^{-\gamma} - 1$ 的图形呈现高度线性，直接验证了公式的预测。三个数据集规模的图形几乎重合，进一步确认了 γ 的稳定性。

Scaling Law

①

- Q Chinchilla Scaling Law
- Fisher +
- $D_{\text{eff}} = D \cdot Q^\gamma$ " =
- " =
- Δ Chen Chang Goyal

②

- $2 \times 3 \times 7 \times 3 = 126$
- Q
- $\Delta L(Q) \propto Q^{(-\gamma)-1}$ —
- Scaling $\beta \propto \gamma$

③

- iso-loss " Q D "
-
- $(\quad / \quad / \quad)$ —
- Q

" "

Scaling Law

Scaling Law

" → "

Q

Y

Huber Loss

$$L(N,D,Q) \rightarrow$$



• Y

• Q

Q

